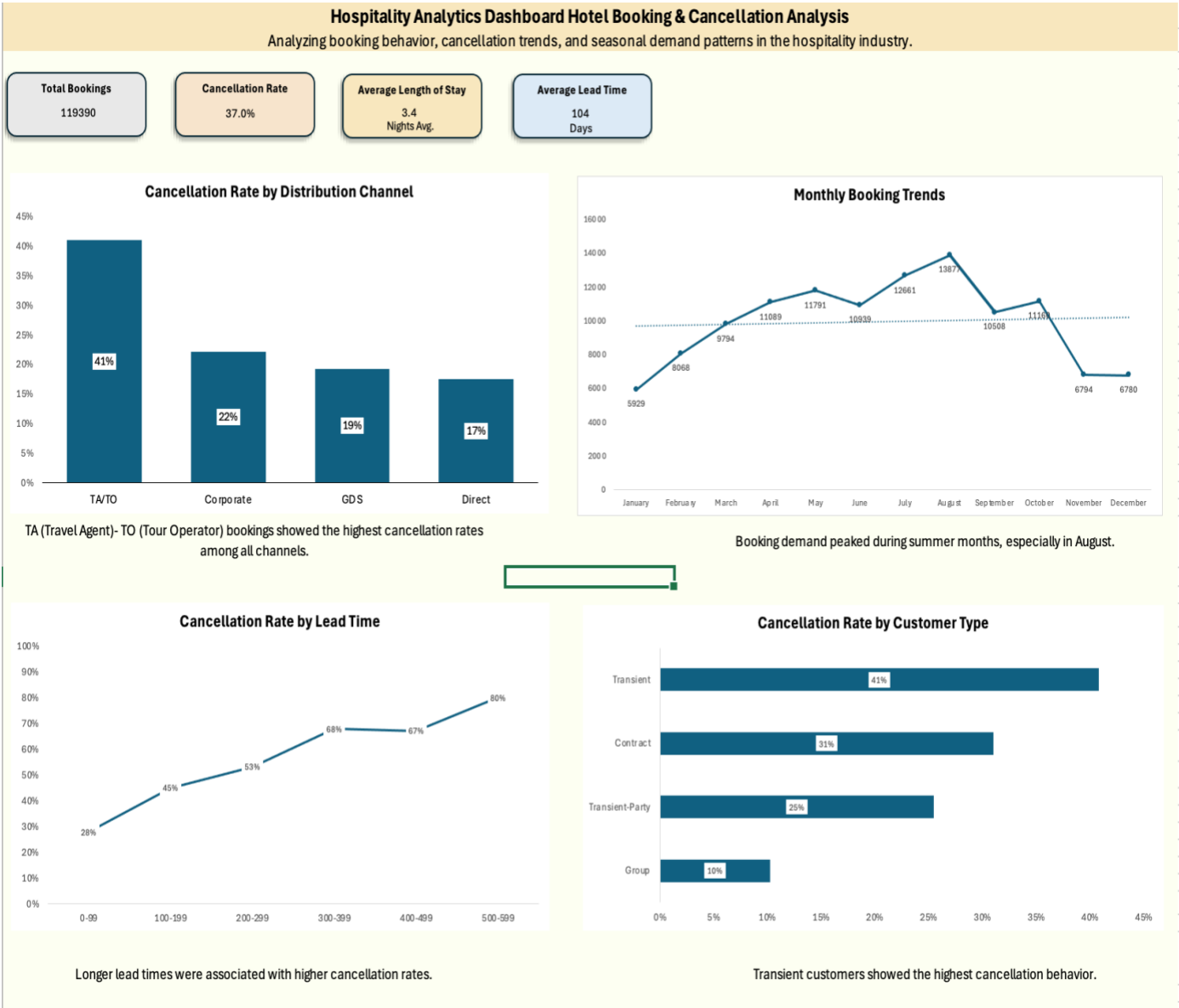


Hotel Booking & Cancellation Analysis

Business Problem

Hotels often face operational and financial challenges, related to unpredictable customer behavior, particularly booking cancellations and seasonal demand fluctuations.

The objective of this project was to identify cancellation patterns, customer trends, and operational insights that could support better forecasting, staffing, and revenue management decisions.



Key Findings

- TA/TO bookings had the highest cancellation rate (41%), while Direct bookings were the most reliable channel (17%).
- Booking demand peaked in August (13,877 reservations), revealing clear seasonal trends.
- Transient customers represented the highest-risk customer segment for cancellations (41%).
- Group bookings showed the most stable booking behavior, with only a 10% cancellation rate.
- Longer lead times were strongly associated with increased cancellation risk, reaching 80% for reservations made 500–599 days in advance.

Tools Used

- Data cleaning
- Excel
- Pivot Tables
- KPI Dashboards
- Data Visualization
- Business Analytics

Business Recommendations

Based on these findings, hotels should:

- Prioritize direct booking channels through loyalty programs and direct-booking incentives.
- Apply stricter cancellation policies or deposit requirements for reservations with long lead times.
- Develop targeted strategies for high-risk customer segments, particularly Transient and TA/TO bookings.
- Use seasonal demand patterns to optimize staffing levels, pricing strategies, and inventory management.
- Implement controlled overbooking strategies during periods and segments with historically high cancellation rates.